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Problem setup

$\dot{x} = -0.1 yz, \quad \dot{y} = 1.6 xz, \quad \dot{z} = -0.75 xy, \quad u_0 = (0.5, 0.75, 1.0)$
 $t \in [0, 13.26446] \cdot 27 \text{ windows} \cdot 1,536 \text{ points/window} \cdot \text{float64}$

REPEAT FOR EACH WINDOW $k = 1 \dots 27$

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PINN for window k

Input

t

Hidden

$4 \times 60 \text{ tanh}$

Output

$N_k(t) \in \mathbb{R}^3$

Hard IC

$\hat{u}_k = u_{0,k} + (t - t_k) N_k(t)$

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ODE residual

$r_k = d\hat{u}_k/dt - f(\hat{u}_k)$, derivatives by autodiff

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Causal residual loss

$\mathcal{L}_i = \text{mean}(r_i^2), \quad w_i = \exp\left(-\varepsilon \sum_{j < i} \mathcal{L}_j\right), \quad \mathcal{L} = \text{mean}(w_i \mathcal{L}_i)$
 $\varepsilon : 0.01 \rightarrow 0.1 \rightarrow 1 \rightarrow 10$

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Adam optimization

learning rate $\times 0.9$ every 1,000 steps

Stage complete?

$\min w > 0.99$
or 4,000 steps

NO

YES

YES

All ε values used?

NO

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L-BFGS

$\leq 1,000$ iterations
unweighted residual MSE

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Save window

checkpoint $\cdot$ history $\cdot$ prediction

Last window?

$k = 27$

NO

YES

Next window

$u_{0,k+1} = \hat{u}_k(t_{k+1})$
copy weights, $k \leftarrow k + 1$

next window

next ε

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Reference evaluation

DOP853 $\cdot$ 13,265 points $\cdot$ rtol 10^{-10} $\cdot$ atol 10^{-12}
used only for evaluation, never in training

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Evaluation outputs

MAE $\cdot$ RMSE $\cdot$ max error $\cdot$ residual MSE $\cdot$ invariant drift $\cdot$ training cost